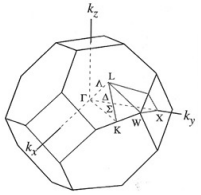
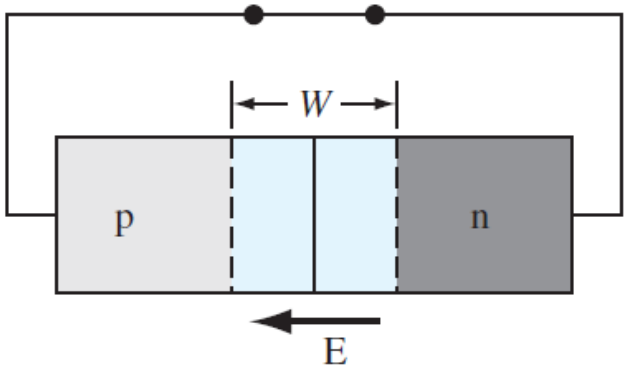
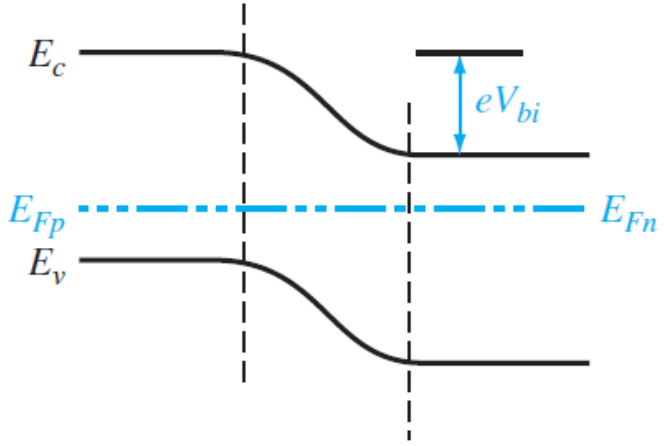
I-V Characteristics of PN Diodes

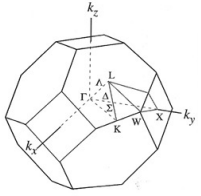


Qualitative Analysis

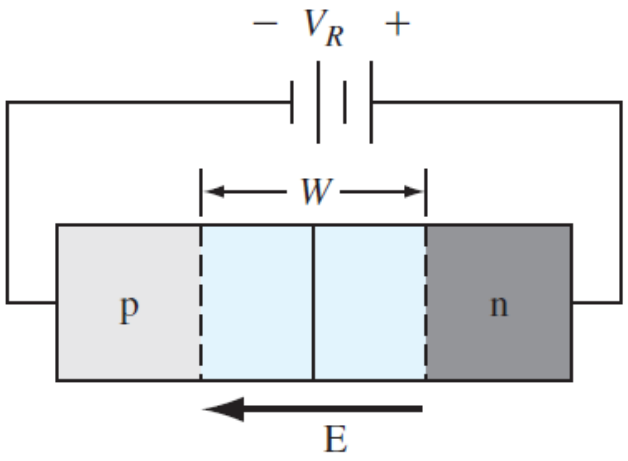
- In Equilibrium

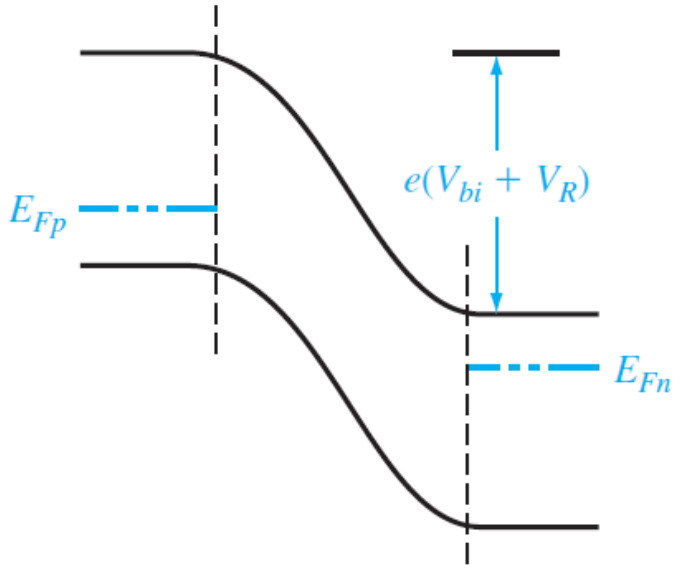


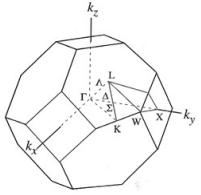




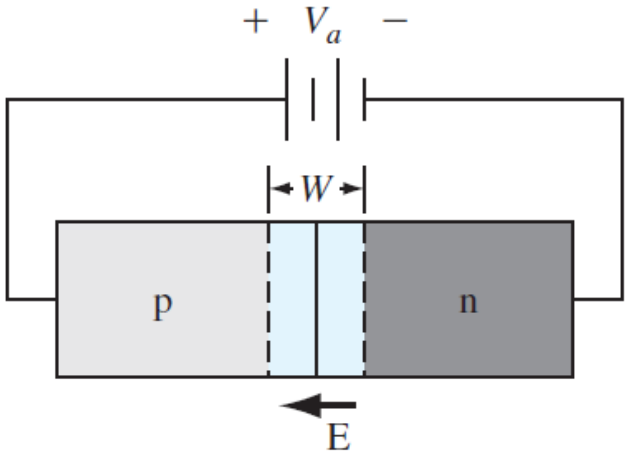
Reverse Bias

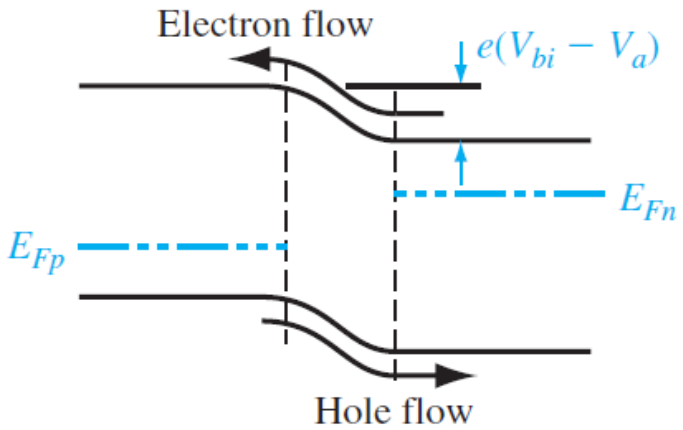






Forward Bias





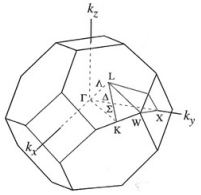
$J_e(V) = J_{r\to l}(V) - J_{l\to r}(V)$

$J_e(0) = J_{r\to l}(0) - J_{l\to r}(0) = 0$

$V \neq 0$. $J_{r\to l}(V) = J_{r\to l}(0)\cdot e^{\frac{eV}{kT}}$

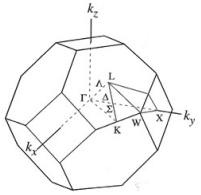
$J_{l\to r}(V) = J_{l\to r}(0)$

$\Rightarrow J_e(V) = J_{r\to l}(0)\cdot\left[e^{\frac{eV}{kT}} - 1\right]$



Ideal Current-Voltage Relationship

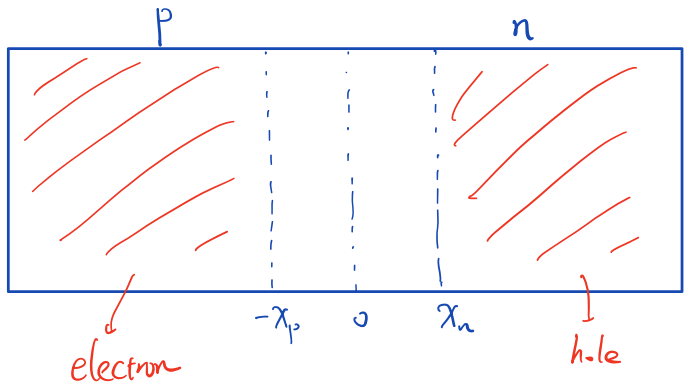
- Abrupt depletion layer approximation
- Maxwell-Boltzmann approximation
- Low injection and complete ionization
- Total current is a constant throughout the entire PN structure
- No generation and recombination in the depletion region
  - ➢The individual electron and hole currents are constant throughout the depletion region

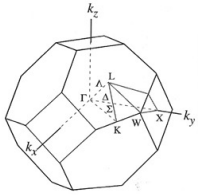


How Can We Derive the I-V?

- What tool/equation do we have?
- $\frac{\partial p}{\partial t} = -\frac{1}{e}\nabla \cdot J_q + g_p - r_p$
- $\frac{\partial n}{\partial t} = \frac{1}{e}\nabla \cdot J_n + g_n - r_n$
- $J_p = ep\mu_p \vec{E} - eD_p \nabla p$
- $J_n = en\mu_n \vec{E} + eD_n \nabla n$
- $\nabla \cdot D = q(p - n + N_d^+ - N_a^-)$

track minority carrier in Quasi nutral area





Track Minority Carrier

- For p-type semiconductor:
  - $\frac{\partial \delta n_p}{\partial t} = D_n \frac{\partial^2 \delta n_p}{\partial x^2} + g_n - r_n$
  - $g_n - r_n = -\frac{\delta n_p}{\tau_n}$

$\frac{\partial \delta n_p}{\partial t} = 0 \Rightarrow D_n \cdot \frac{\partial^2 \delta n_p}{\partial x^2} - \frac{\delta n_p}{\tau_n} = 0$

$L_n^2 = D_n \cdot \tau_n \quad \frac{\partial^2 \delta n_p}{\partial x^2} - \frac{\delta n_p}{L_n^2} = 0$

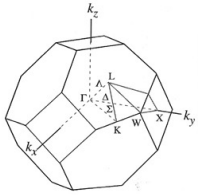
- For n-type semiconductor:
  - $\frac{\partial \delta p_n}{\partial t} = D_p \frac{\partial^2 \delta p_n}{\partial x^2} + g_p - r_p$
  - $g_p - r_p = -\frac{\delta p_n}{\tau_p}$

$\Rightarrow \delta n_p = A e^{\frac{x}{L_n}} + B \cdot e^{-\frac{x}{L_n}}$

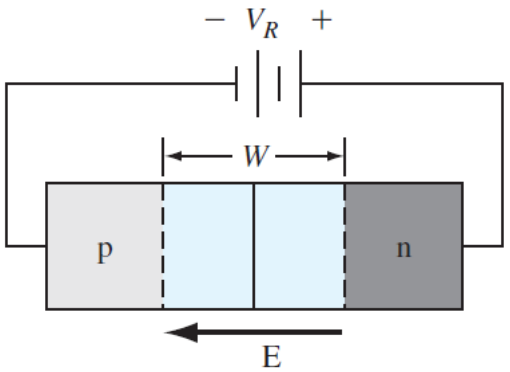
$B=0 \Rightarrow \delta n_p = A \cdot e^{\frac{x}{L_n}}$

$\delta n_p(x) = \delta n_p(-x_p) \cdot e^{\frac{x+x_p}{L_n}} \quad x \le -x_p$

$\delta p_n(x) = \delta p_n(x_n) \cdot e^{-\frac{x-x_n}{L_p}} \quad x \ge x_n$



We Can Solve these Equations

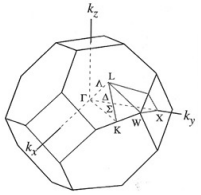


$$\frac{d^2(\delta p_n)}{dx^2} - \frac{\delta p_n}{L_p^2} = 0 \qquad (x > x_n)$$

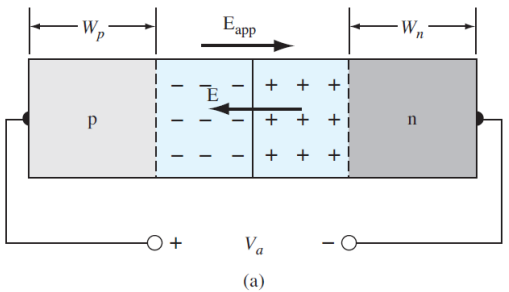
$$\frac{d^2(\delta n_p)}{dx^2} - \frac{\delta n_p}{L_n^2} = 0 \qquad (x < x_p)$$

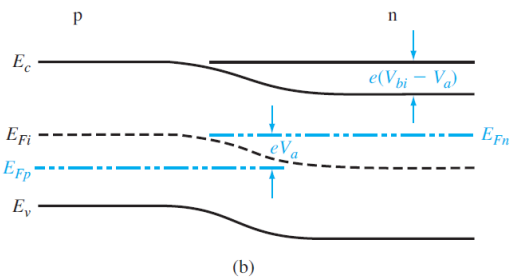
$$\delta p_n(x) = p_n(x) - p_{n0} = Ae^{x/L_p} + Be^{-x/L_p} \qquad (x \geq x_n)$$

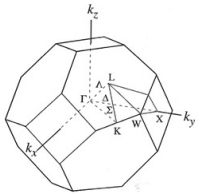
$$\delta n_p(x) = n_p(x) - n_{p0} = Ce^{x/L_n} + De^{-x/L_n} \qquad (x \leq -x_p)$$



Now We Need to Fix the Boundary Condition

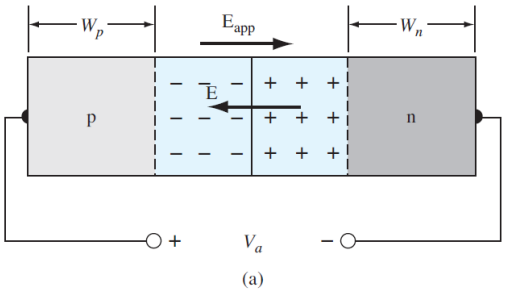






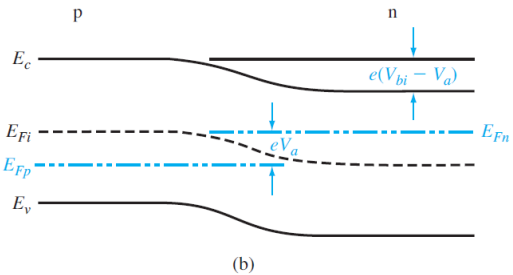
- Where is the boundary?
- What is the boundary condition?
  - ➢ What is the boundary condition at zero bias?

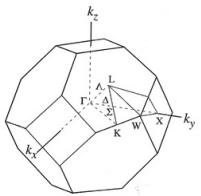
Now We Need to Fix the Boundary Condition



$$n_p = n_{p0} \exp\left(\frac{eV_a}{kT}\right)$$

$$p_n = p_{n0} \exp\left(\frac{eV_a}{kT}\right)$$





Boundary Condition

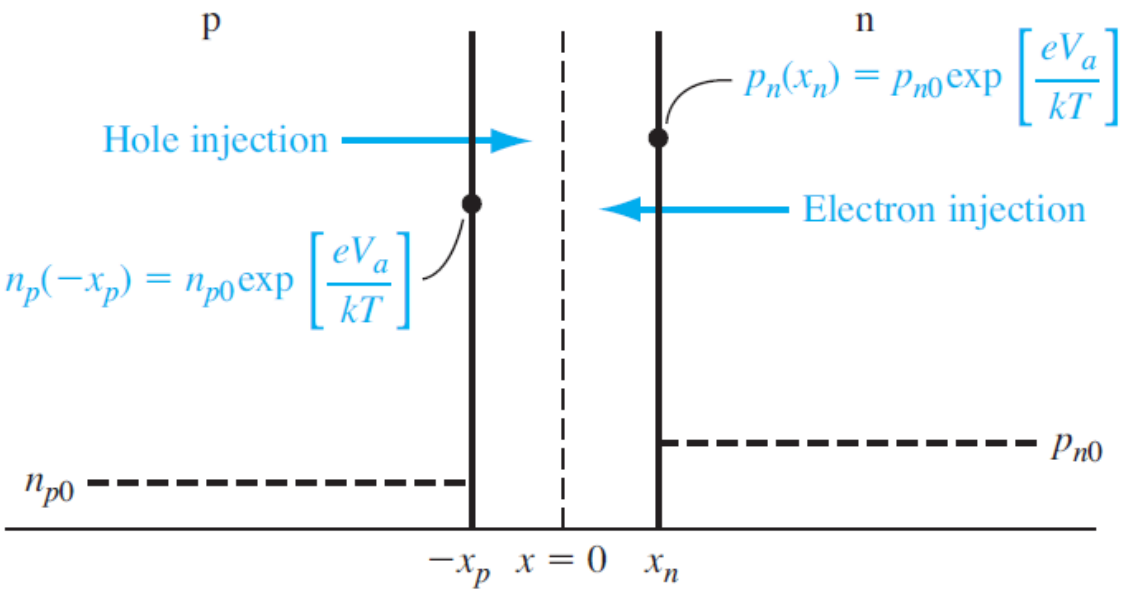
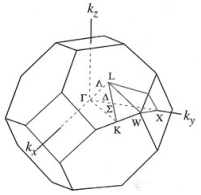


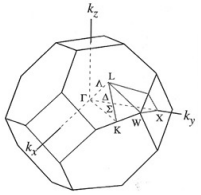
Figure 8.4 | Excess minority carrier concentrations at the space charge edges generated by the forward-bias voltage.



Excess Carrier Concentrations

$$\delta p_n(x) = p_n(x) - p_{n0} = p_{n0}\left[\exp\left(\frac{eV_a}{kT}\right) - 1\right]\exp\left(\frac{x_n - x}{L_p}\right) \qquad (x \geq x_n)$$

$$\delta n_p(x) = n_p(x) - n_{p0} = n_{p0}\left[\exp\left(\frac{eV_a}{kT}\right) - 1\right]\exp\left(\frac{x_p + x}{L_n}\right) \qquad (x \leq -x_p)$$



Excess Carrier Concentrations

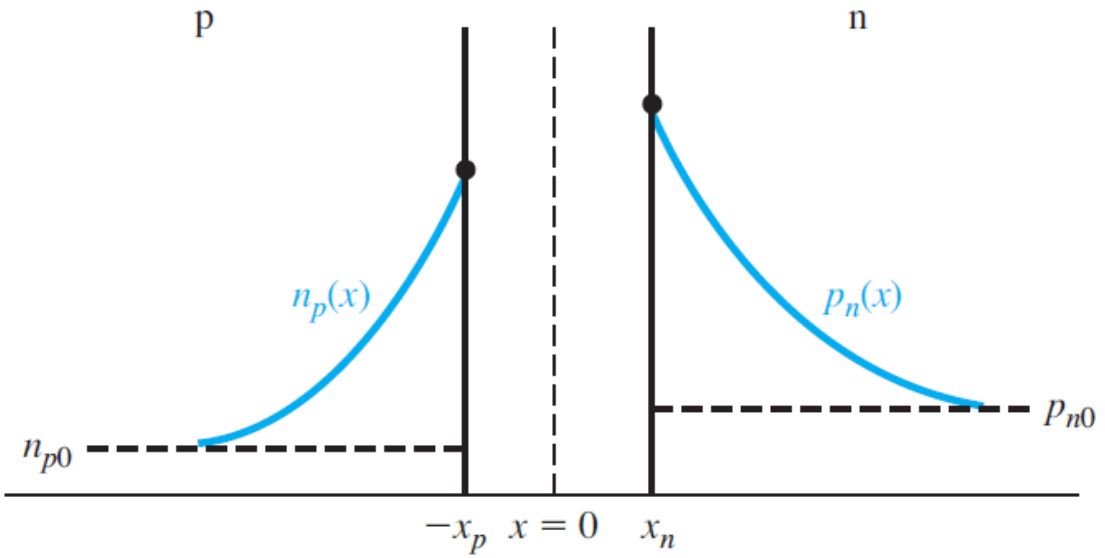
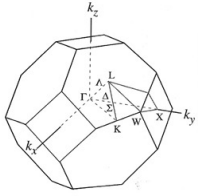


Figure 8.5 | Steady-state minority carrier concentrations in a pn junction under forward bias.



Quasi-Fermi Levels

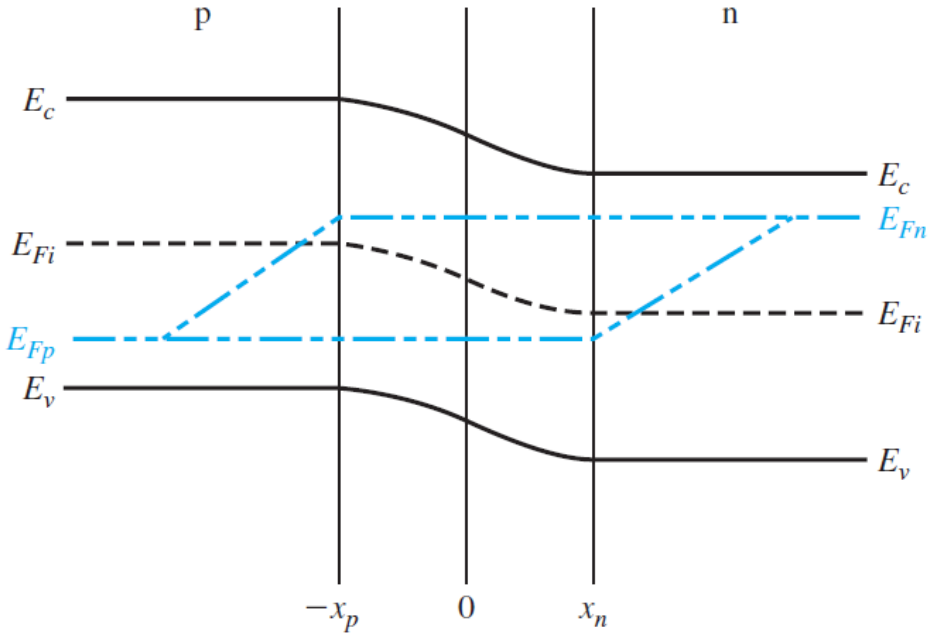
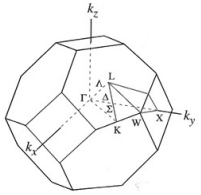


Figure 8.6 | Quasi-Fermi levels through a forward-biased pn junction.

$$np = n_i^2 \exp\left(\frac{E_{Fn} - E_{Fp}}{kT}\right)$$

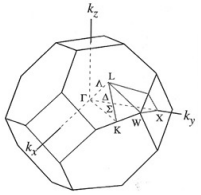


Ideal pn Junction Current

$$J_p(x_n) = -eD_p \left.\frac{dp_n(x)}{dx}\right|_{x=x_n}$$

$$J_p(x_n) = \frac{eD_p p_{n0}}{L_p}\left[\exp\left(\frac{eV_a}{kT}\right) - 1\right]$$

$$J_n(-x_p) = \frac{eD_n n_{p0}}{L_n}\left[\exp\left(\frac{eV_a}{kT}\right) - 1\right]$$



Ideal pn Junction Current

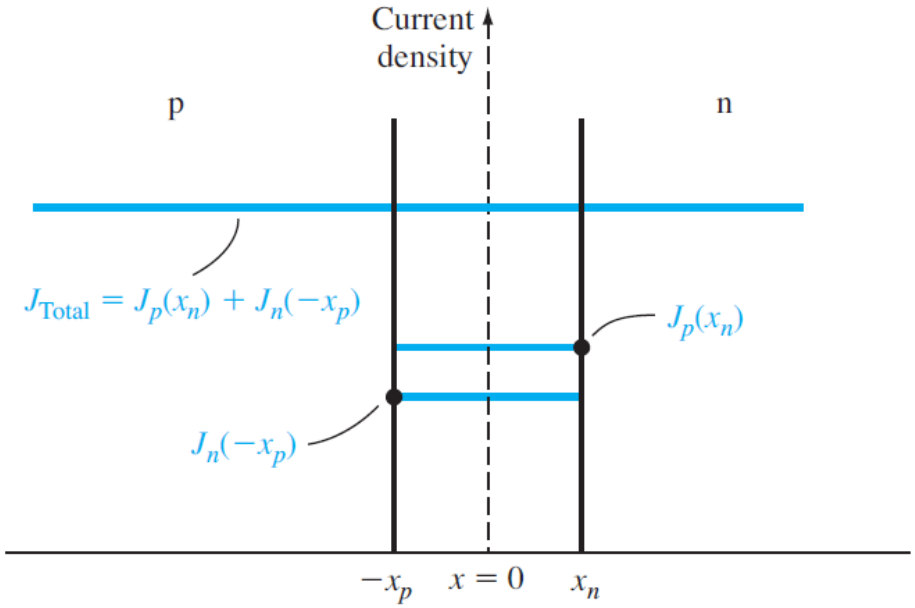
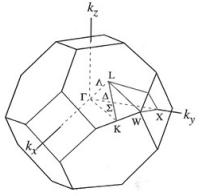


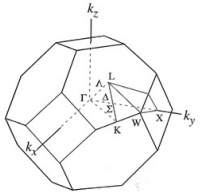
Figure 8.7 | Electron and hole current densities through the space charge region of a pn junction.



Ideal pn Junction Current

$$J = J_p(x_n) + J_n(-x_p) = \left[\frac{eD_p p_{n0}}{L_p} + \frac{eD_n n_{p0}}{L_n}\right]\left[\exp\left(\frac{eV_a}{kT}\right) - 1\right]$$

$$J = J_s\left[\exp\left(\frac{eV_a}{kT}\right) - 1\right]$$



Ideal pn Junction Current

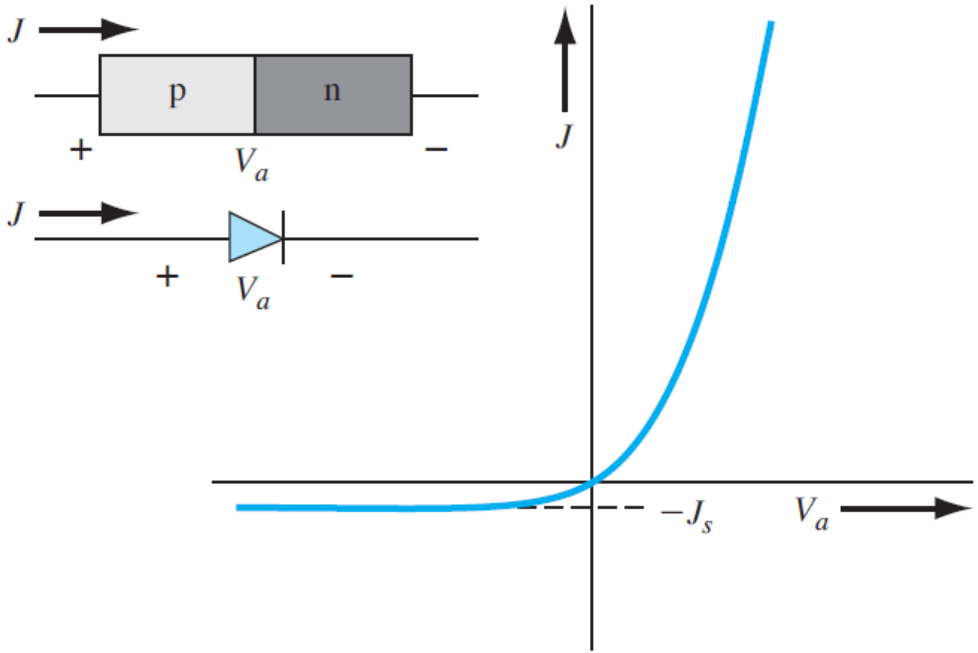
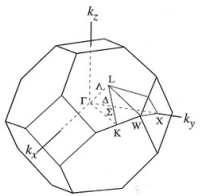


Figure 8.8 | Ideal *I–V* characteristic of a pn junction diode.



Summary of the Ideal I-V Characteristics

$$J_p(x) = \frac{eD_p p_{n0}}{L_p}\left[\exp\left(\frac{eV_a}{kT}\right) - 1\right]\exp\left(\frac{x_n - x}{L_p}\right) \qquad (x \geq x_n)$$

$$J_n(x) = \frac{eD_n n_{p0}}{L_n}\left[\exp\left(\frac{eV_a}{kT}\right) - 1\right]\exp\left(\frac{x_p + x}{L_n}\right) \qquad (x \leq -x_p)$$

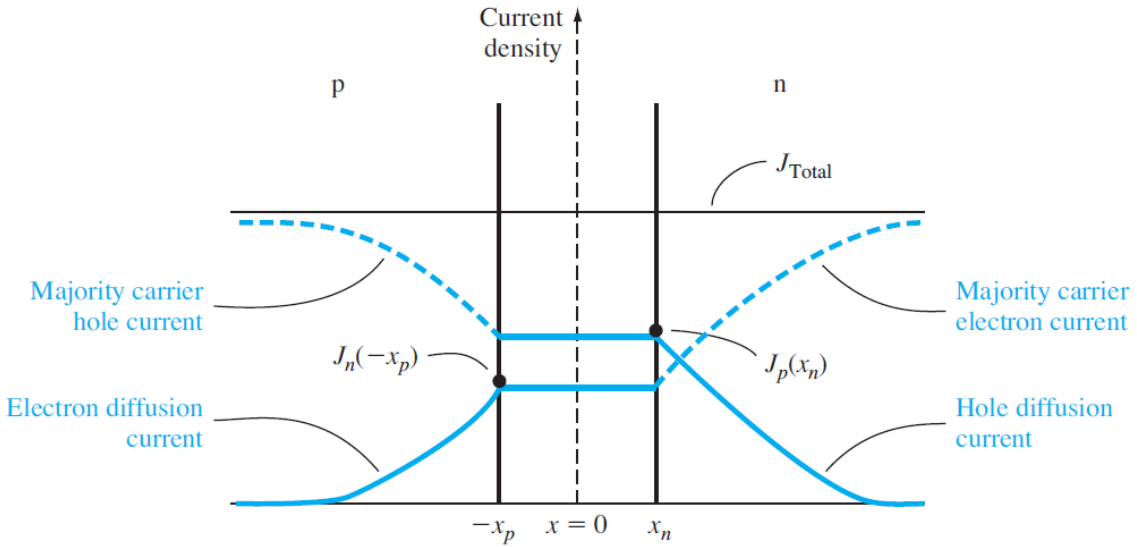
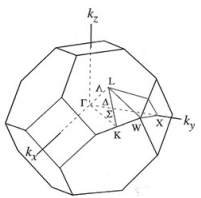


Figure 8.10 | Ideal electron and hole current components through a pn junction under forward bias.



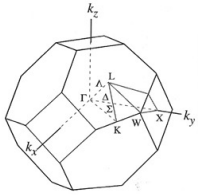
Temperature Effects

- Reverse Bias:

$$J \propto \exp\left(\frac{-E_g}{kT}\right)$$

- Forward Bias

$$J \propto \exp\left(\frac{-E_g}{kT}\right)\exp\left(\frac{eV_a}{kT}\right)$$



One Special Case: Short Diode

$$p_n(x_n) = p_{n0} \exp\left(\frac{eV_a}{kT}\right)$$

$$p_n(x = x_n + W_n) = p_{n0}$$

Infinite surface-recombination velocity

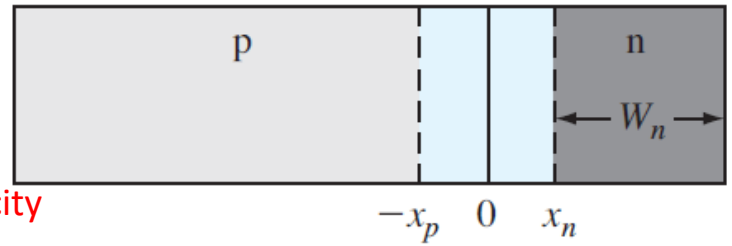
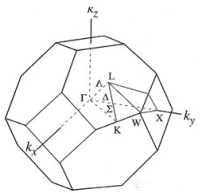


Figure 8.11 | Geometry of a "short" diode.

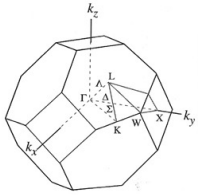
$$\delta p_n(x) = p_{n0}\left[\exp\left(\frac{eV_a}{kT}\right) - 1\right]\frac{\sinh\left[(x_n + W_n - x)/L_p\right]}{\sinh[W_n/L_p]}$$

$$\delta p_n(x) = p_{n0}\left[\exp\left(\frac{eV_a}{kT}\right) - 1\right]\left(\frac{x_n + W_n - x}{W_n}\right)$$

$$J_p(x) = \frac{eD_p p_{n0}}{W_n}\left[\exp\left(\frac{eV_a}{kT}\right) - 1\right]$$

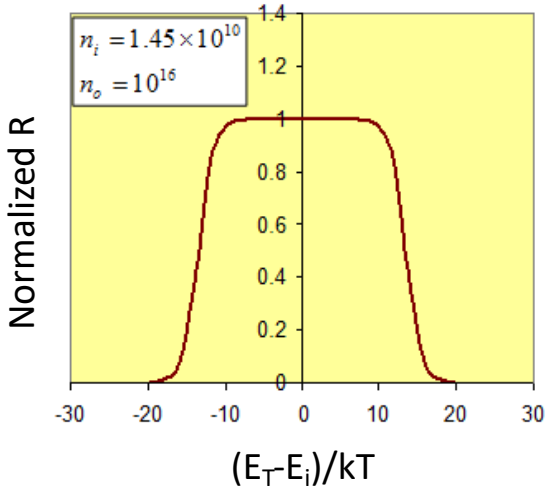


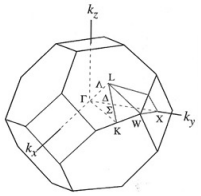
Generation-Recombination Current



Net Generation-Recombination Rate

- $R = \dfrac{(pn - n_i^2)}{\tau_p\left(n + n_i \exp\left(\frac{E_T - E_i}{kT}\right)\right) + \tau_n\left(p + n_i \exp\left(\frac{E_i - E_T}{kT}\right)\right)}$





Reverse Bias Generation Current

- $R = \dfrac{(pn - n_i^2)}{\tau_p\left(n + n_i \exp\left(\frac{E_T - E_i}{kT}\right)\right) + \tau_n\left(p + n_i \exp\left(\frac{E_i - E_T}{kT}\right)\right)}$
- $n \approx p \approx 0$
- $R = \dfrac{(-n_i^2)}{\tau_p\left(n_i \exp\left(\frac{E_T - E_i}{kT}\right)\right) + \tau_n\left(n_i \exp\left(\frac{E_i - E_T}{kT}\right)\right)}$
  - If traps energy is close to mid-gap, $R = \dfrac{-n_i}{(\tau_p + \tau_n)} = \dfrac{-n_i}{2\tau} = -G$
  - So: $J_{gen} = \int_0^W e\, G dx$

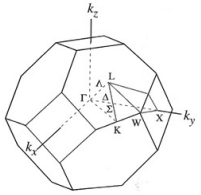
$$J_{gen} = \frac{e n_i W}{2\tau_0} \quad \propto n_i \propto \exp\left(-\frac{E_g}{2kT}\right)$$

Reverse: diffuse dominate $\propto n_i^2 \propto \exp\left(\frac{E_g}{kT}\right)$

G & R dominate $\propto n_i \propto \exp\left(\frac{E_g}{2kT}\right)$

Forward: diffuse dominate $\propto n_i^2 \cdot \exp\left(\frac{eV}{kT}\right)$

G & R dominate $\propto n_i \cdot \exp\left(\frac{eV}{2kT}\right)$



Forward Bias Generation Current

- $R = \dfrac{(pn - n_i^2)}{\tau_p\left(n + n_i \exp\left(\frac{E_T - E_i}{kT}\right)\right) + \tau_n\left(p + n_i \exp\left(\frac{E_T - E_i}{kT}\right)\right)}$

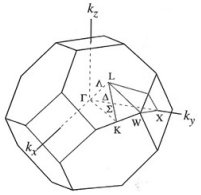
$$n = n_i \exp\left(\frac{eV_a}{2kT}\right)$$

$$p = n_i \exp\left(\frac{eV_a}{2kT}\right)$$

$$R_{\max} = \frac{n_i}{2\tau_0} \frac{[\exp(eV_a/kT) - 1]}{[\exp(eV_a/2kT) + 1]}$$

$$R_{\max} = \frac{n_i}{2\tau_0} \exp\left(\frac{eV_a}{2kT}\right)$$

$$J_{\text{rec}} = \frac{eWn_i}{2\tau_0} \exp\left(\frac{eV_a}{2kT}\right) = J_{r0} \exp\left(\frac{eV_a}{2kT}\right)$$



Total Forward Bias Current

$$J_{\text{rec}} = \frac{eWn_i}{2\tau_0}\exp\left(\frac{eV_a}{2kT}\right) = J_{r0}\exp\left(\frac{eV_a}{2kT}\right)$$

$$\ln J_{\text{rec}} = \ln J_{r0} + \frac{eV_a}{2kT} = \ln J_{r0} + \frac{V_a}{2V_t}$$

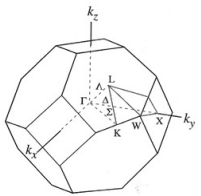
$$J_D = J_s\exp\left(\frac{eV_a}{kT}\right)$$

$$\ln J_D = \ln J_s + \frac{eV_a}{kT} = \ln J_s + \frac{V_a}{V_t}$$

$$J = J_{\text{rec}} + J_D$$

$$I = I_s\left[\exp\left(\frac{eV_a}{nkT}\right) - 1\right]$$

n=1 for diffusion dominated
n=2 for recombination dominated



Series Resistance

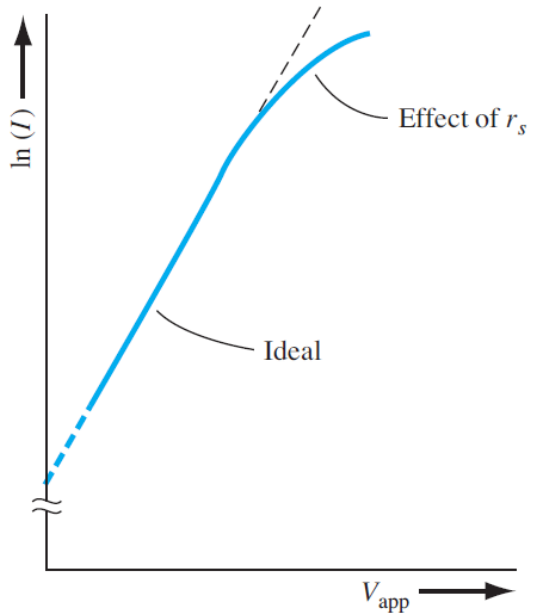
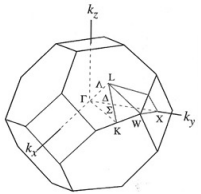


Figure 8.23 | Forward-biased *I–V* characteristics of a pn junction diode showing the effect of series resistance.



Junction Breakdown

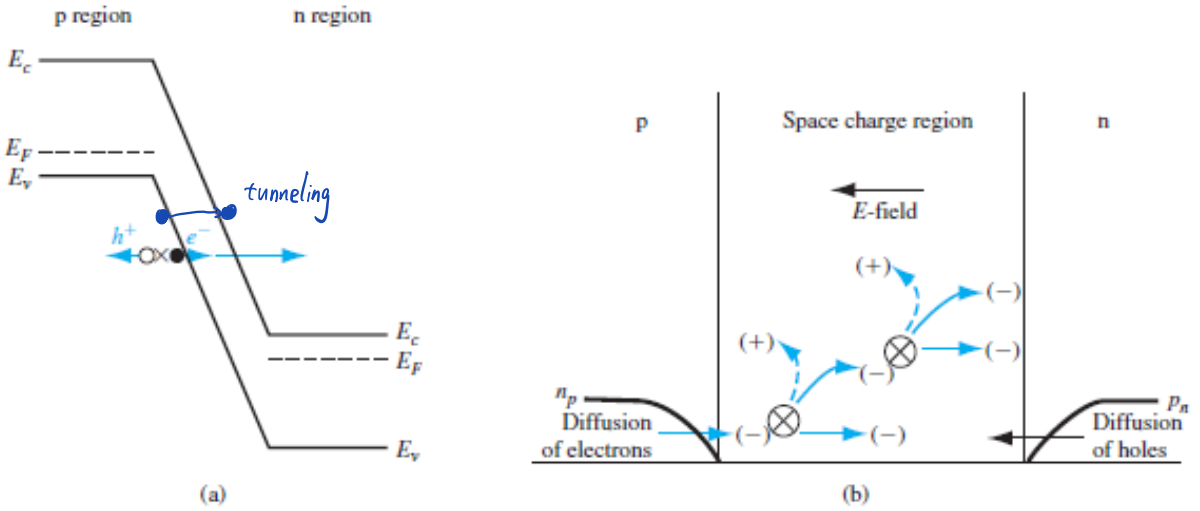
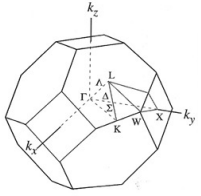
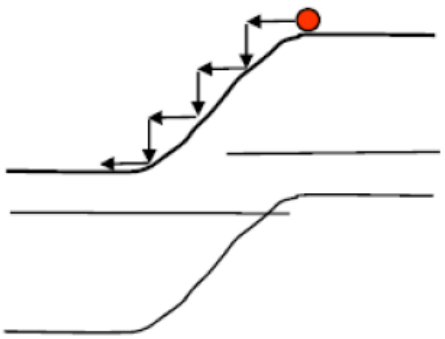


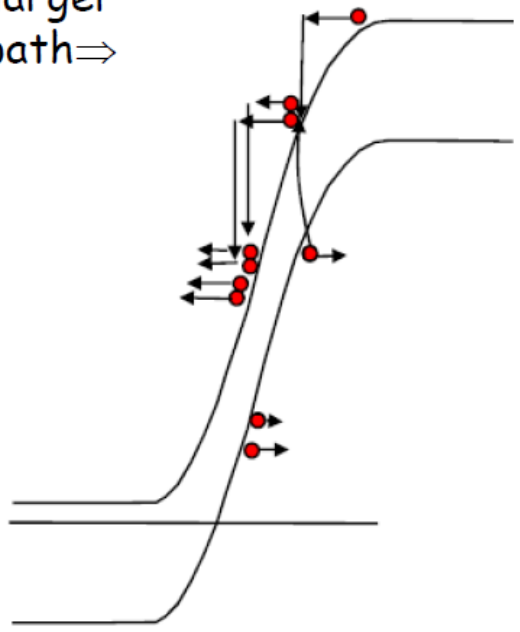
Figure 7.12 | (a) Zener breakdown mechanism in a reverse-biased pn junction; (b) avalanche breakdown process in a reverse-biased pn junction.

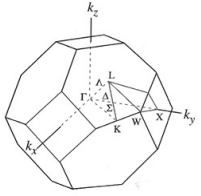


Impact Ionization

Depletion width larger than mean free path⇒ lots of collisions

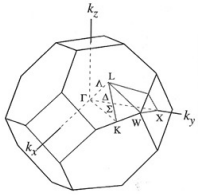






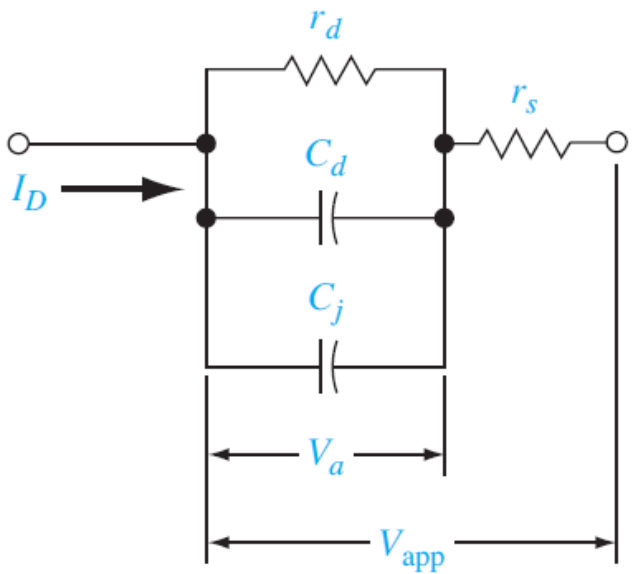
Junction Breakdown

- Impact ionization is more noisy than Zener tunneling
  - ➢Why?
- Impact ionization is more temperature sensitive
  - ➢Why?



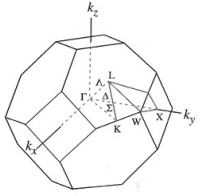
Small-Signal Model of PN Junction

Equivalent Circuit



C_d: diffusion capacitance

C_j: junction capacitance.



Diffusion Resistance

$$g_d = \left.\frac{dI_D}{dV_a}\right|_{V_a=V_0} = \left(\frac{e}{kT}\right) I_s \exp\left(\frac{eV_0}{kT}\right) \approx \frac{I_{DQ}}{V_t}$$

$$r_d = \frac{V_t}{I_{DQ}}$$

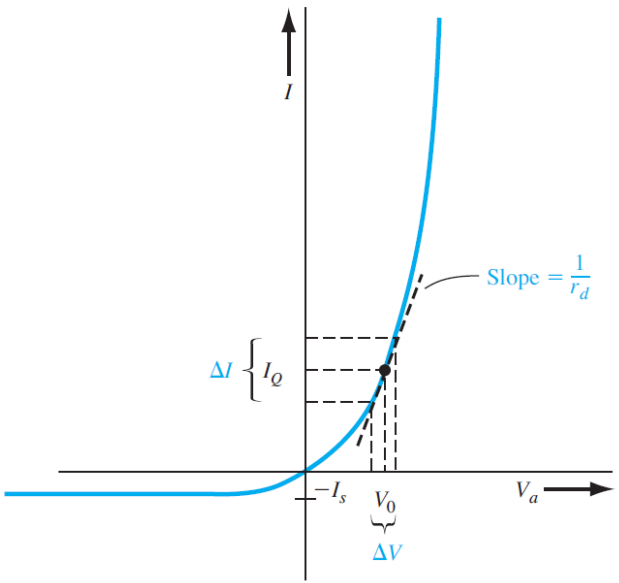
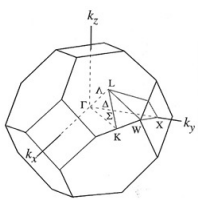
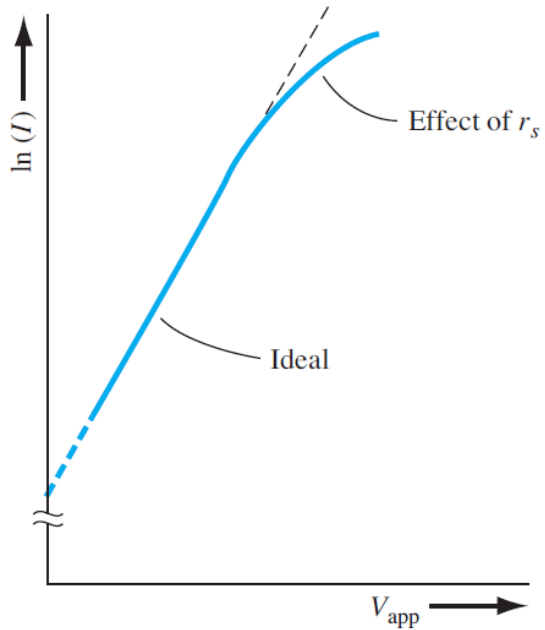


Figure 8.18 | Curve showing the concept of the small-signal diffusion resistance.

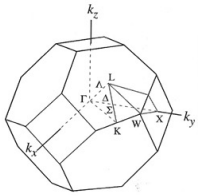


Contact Resistance



$$V_{app} = V_a + Ir_s$$

Figure 8.23 | Forward-biased *I–V* characteristics of a pn junction diode showing the effect of series resistance.



Diffusion Capacitance

